RESULTS

Results

01 · CB-SEM

confirmatory structural model

Table 1. Measurement model (loadings, reliability & item descriptives)

Construct / Item	Mean	SD	B	SE	z	p	Std. loading	ω	α	CR	AVE
visual								*.61*	*.63*	*.61*	*.37*
x1	4.94	1.17	1.00	0.00	0	<.001	.77				
x2	6.09	1.18	0.55	0.14	3.97	<.001	.42				
x3	2.25	1.13	0.73	0.15	4.85	<.001	.58				
textual								*.89*	*.88*	*.89*	*.72*
x4	3.06	1.16	1.00	0.00	0	<.001	.85				
x5	4.34	1.29	1.11	0.07	16.50	<.001	.86				
x6	2.19	1.10	0.93	0.06	14.80	<.001	.84				
speed								*.69*	*.69*	*.69*	*.42*
x7	4.19	1.09	1.00	0.00	0	<.001	.57				
x8	5.53	1.01	1.18	0.15	7.90	<.001	.72				
x9	5.37	1.01	1.08	0.38	2.82	.005	.67				

Item sample: Item Mean/SD are computed on the listwise estimation sample (the same N as the model fit).

Table 2. Discriminant validity (Fornell–Larcker)

	visual	textual	speed
visual	*.61*		
textual	.46***	*.85*	
speed	.47***	.28***	*.65*

*p<.05, **p<.01, ***p<.001

Table 3. Discriminant validity (HTMT)

	visual	textual	speed
visual			
textual	.38		
speed	.39	.28	

Discriminant validity: Discriminant validity also has its own card (AVE / convergent validity); it is included here so one run gives the complete measurement-model writeup.

Table 4. Fit indices

χ^2 (df, p)	χ^2/df	CFI	TLI	RMSEA [90% CI]	SRMR
85.31 (24, <.001)	3.55	.93	.90	.09 [.07, .11]	.07

Table 5. Structural paths, indirect effects & moderation

H	Path	B	Std. β	p	Percentile 95% CI		BC 95% CI		Result
					Lower	Upper	Lower	Upper	
Direct paths									
H1	visual → textual	0.50	.46	<.001	.32	.70	.33	.72	Supported
H2	textual → speed	0.05	.09	.333	−.05	.17	−.04	.17	Not supported
H3	visual → speed	0.30	.43	<.001	.14	.45	.16	.48	Supported
Indirect effects									
H4	visual → textual → speed	0.03	.04	.346	−.03	.09	−.02	.10	Not supported

Indirect effects: The indirect-effects section of Table 5 appears only when the drawn structural paths form a chain (X → M → Y); each indirect effect is a lavaan defined effect with a bootstrapped 95% CI.

CIs: Bias-corrected CIs benefit from ≥7,000 resamples (Andrews & Buchinsky, 2000); consider the 10,000 publication-grade preset for final runs.

Scope: Tables shown follow the pipeline stages you ran (EFA → CFA → fit → structural); if EFA was deselected, the E1/E2 preamble is omitted; if the structural stage was deselected, Table 5 is omitted.

Cutoffs: Good-fit guidelines (Hu & Bentler, 1999; Marsh, Hau & Wen, 2004): CFI/TLI ≥ .95, RMSEA ≤ .06 [90% CI], SRMR ≤ .08 — guidelines, not pass/fail gates; RMSEA is unstable at small df / small N, so interpret it cautiously for compact models.

Estimator: Use WLSMV for ordinal indicators.

R^2: R^2 is filled once per endogenous (outcome) construct. R^2(textual) = .21, R^2(speed) = .23

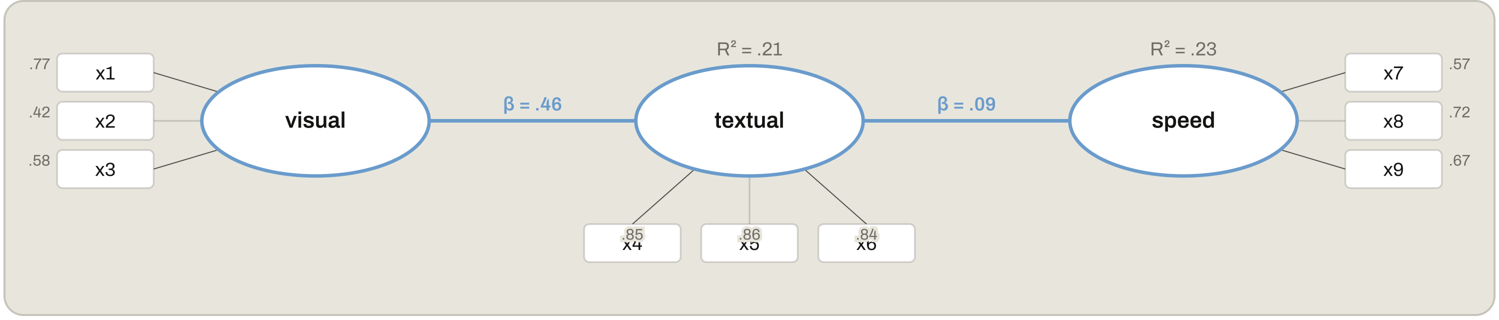
Caution: EFA on the same sample is exploratory — treat it as a diagnostic, not confirmatory evidence.

Moderation: Moderation edges appear only when drawn on the canvas, each an interaction-term effect from the same bootstrap run; simple slopes at -1 SD / mean / +1 SD are estimated as defined parameters.

Figure. Model

– + Fit

Draw path Move Delete



Understanding the numbers

CFI. How much better the model fits than a baseline with no relationships at all (≥ .95 is a common, non-binding guideline). Here, CFI = .93.

RMSEA. The average model misfit per degree of freedom, penalizing complexity (≤ .06 is a common, non-binding guideline). Here, RMSEA = .09 [90% CI .07, .11].

TLI. Like CFI, compares the model to a baseline with no relationships, penalizing complexity more than CFI does (≥ .95 is a common, non-binding guideline). Here, TLI = .90.

SRMR. The average standardized difference between the observed and model-implied correlations (≤ .08 is a common, non-binding guideline). Here, SRMR = .07.

χ^2 (model fit). Tests whether the model-implied covariances differ from the observed ones; RMSEA is unstable at small df/small N, so this and the other fit indices are read together, not in isolation. Here, $\chi^2(24) = 85.31$, $p < .001$.

χ^2/df. The chi-square divided by its degrees of freedom - a size-adjusted view of the same model-fit discrepancy. Here, $\chi^2/df = 3.55$.

Mean. An item's own observed average, shown for descriptive context alongside its loading. Here, item means range from 2.19 to 6.09 across the 9 items in the measurement-model table.

SD. An item's own observed standard deviation. Here, item SDs range from 1.01 to 1.29 across the 9 items above.

B. The unstandardized estimate for whichever row you're reading - a factor loading, a structural path coefficient, or (under moderation) a simple slope. Here, factor-loading B's range from 0.55 to 1.18 across the 9 items; the structural-paths and conditional-effects tables each report their own B per row.

SE. The standard error of the estimate in whichever row you're reading - a factor loading or a simple slope. Here, factor-loading SEs range from 0.00 to 0.38 across the 9 items; the conditional-effects table reports its own SE per moderator level.

z. A loading's estimate divided by its SE - how many standard errors it is from zero. Here, loading z-values range from 0 to 16.50 across the 9 items.

p. The significance of the estimate in whichever row you're reading - a loading, a structural path, or a moderation edge. Here, factor-loading p-values range from <.001 to .005 across items, and structural-path p-values range from <.001 to .346; the merged table's own Result column is instead derived from whether the bootstrap CI excludes zero, not from p.

Std. loading. The loading rescaled so item variances are comparable across the model. Here, standardized loadings range from .42 to .86 across the 9 items.

ω (omega). Model-based reliability for a construct, reported once per construct alongside its loadings. Here, ω ranges from .61 to .89 across the 3 constructs.

α (alpha). Cronbach's alpha for a construct, reported alongside ω. Here, α ranges from .63 to .88 across the 3 constructs.

CR. Composite reliability for a construct - part of confirming the measurement model before reading the structural paths (this card's own how-to-read). Here, CR ranges from .61 to .89 across the 3 constructs.

AVE. Average variance extracted for a construct - part of the same CR/AVE adequacy check. Here, AVE ranges from .37 to .72 across the 3 constructs.

H. Which hypothesis number a row of the merged table reports - direct paths first, then indirect effects, then moderation, each numbered in the order you drew or defined them. Here, this run reports 4 H-numbered rows.

Std. β. The standardized structural-path coefficient, comparable across paths regardless of each variable's own units. Here, standardized path coefficients range from .04 to .46 across the 4 rows.

Percentile 95% CI - lower bound. The lower bound of the primary bootstrap confidence interval for a row. Here, percentile-CI lower bounds range from −.05 to .32 across the structural paths.

Percentile 95% CI - upper bound. The upper bound of the primary bootstrap confidence interval for a row. Here, percentile-CI upper bounds range from .17 to .70 across the structural paths.

BC 95% CI - lower bound. The lower bound of the bias-corrected bootstrap interval, shown alongside the percentile CI. Here, bias-corrected lower bounds range from −.04 to .33 across the structural paths.

BC 95% CI - upper bound. The upper bound of the bias-corrected bootstrap interval. Here, bias-corrected upper bounds range from .17 to .72 across the structural paths.

Result. Whether a row's percentile 95% bootstrap CI excludes zero (Supported) or straddles it (Not supported), at α = .05. Here, 2 of 4 rows read Supported.

Moderator level. Which point on the moderator's distribution a simple slope is estimated at: -1 SD, the mean, or +1 SD. Here, this run estimates simple slopes across 0 moderation edge(s), each at the three fixed levels (-1 SD, mean, +1 SD).

Factor 1 loading. An item's loading on the first rotated factor from the optional EFA-preamble stage. Here, Factor 1 loadings appear in the E2 preamble table when the EFA stage is run.

Factor 2 loading. An item's loading on the second rotated factor from the optional EFA-preamble stage. Here, Factor 2 loadings appear alongside Factor 1 in the E2 preamble table when the EFA stage is run.

Communality. How much of an item's variance is explained by the retained EFA-preamble factors together. Here, communalities appear in the E2 preamble table when the EFA stage is run.

How to read this test

First confirm the measurement model (loadings high, CR/AVE adequate) and overall fit indices. Then read the structural paths: each std. β with p/CI is a hypothesized relationship between constructs; R^2 shows variance explained in each outcome construct. CB-SEM is confirmatory: the measurement model must be specified from theory a priori — any post-hoc respecification (e.g. from modification indices) is exploratory, must be reported as such, and ideally cross-validated on a fresh sample.

APA template: The model fit well (CFI=.93, RMSEA=.09, SRMR=.07); the path from visual to textual gave β=.46, p < .001.

Statistical basis: Model estimation (lavaan) (Rosseel, Y. (2012). "lavaan: An R Package for Structural Equation Modeling." Journal of Statistical Software, 48(2), 1-36.); Fit-index cutoffs (CFI/TLI ≥ .95, RMSEA ≤ .06, SRMR ≤ .08) (Hu, L., Bentler, P. M. (1999). "Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives." Structural Equation Modeling, 6(1), 1-55.); Cutoffs are guidelines, not pass/fail gates (Marsh, H. W., Hau, K. T., Wen, Z. (2004). "In search of golden rules: Comment on hypothesis-testing approaches to setting cutoff values for fit indices." Structural Equation Modeling, 11(3), 320-341.); RMSEA 90% confidence interval (MacCallum, R. C., Browne, M. W., Sugawara, H. M. (1996). "Power analysis and determination of sample size for covariance structure modeling." Psychological Methods, 1(2), 130-149.); AVE / Fornell-Larcker discriminant validity (Fornell, C., Larcker, D. F. (1981). "Evaluating structural equation models with unobservable variables and measurement error." Journal of Marketing Research, 18(1), 39-50.); Reliability preference (ω over α) (McNeish, D. (2018). "Thanks coefficient alpha, we'll take it from here." Psychological Methods, 23(3), 412-433.); Reporting standard (APA JARS-Quant) (Appelbaum, M., Cooper, H., Kline, R. B., Mayo-Wilson, E., Nezu, A. M., Rao, S. M. (2018). "Journal article reporting standards for quantitative research in psychology." American Psychologist, 73(1), 3-25.); Double-mean-centering for latent moderation (semTools::indProd) (Lin, G. C., Wen, Z., Marsh, H. W., Lin, H. S. (2010). "Structural equation models of latent interactions: Clarification of orthogonalizing and double-mean-centering strategies." Structural Equation Modeling, 17(3), 374-391.); Mean-centering strategy for latent interactions (precursor) (Marsh, H. W., Wen, Z., Hau, K. T. (2004). "Structural equation models of latent interactions: evaluation of alternative estimation strategies and indicator construction." Psychological Methods, 9(3), 275-300.); Simple slopes at -1SD/mean/+1SD (Aiken, L. S., West, S. G. (1991). Multiple Regression: Testing and Interpreting Interactions. Sage.). Full references in the exported CITATIONS.txt.

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